

The measurements collected during the validation study can then be used to define the initial target values and control limits. Subsequently, regularly repeated QC measurements can refine the initial MU estimate. It is now well-established that MU tends to decrease as the method is used routinely and eventually stabilizes around a convergence value, provided that the sources of variability are constant over time.

7.4 Use Interlaboratory Comparison Data

The question of the different laboratory comparison procedures was already addressed in Sections 3.1 and 4.3, discussing precision and trueness, respectively. A proficiency testing scheme (PTS) is a procedure that consists of asking a large group of laboratories to perform a measurement, usually without replicates on a common sample, in most cases also without replicates. To verify their abilities in a specific field of application, laboratories can use their method. This procedure is a requirement for accreditation, and ISO 13528 international standard describes the statistical methods that can be used by proficiency testing organizers to interpret the results.

On the other hand, an interlaboratory study differs from a PTS because it is a study that brings together a group, often limited, of laboratories that are asked to perform replicates on the same sample using the same method. The aim is to evaluate the reproducibility of this method. This type of study could be mandatory to document the part of a standard related to the performance of a method. The ISO 5725 series of standards describes the organization and operation of this kind of interlaboratory analysis. However, the two comparison approaches may overlap, and previous descriptions may accept many extensions.

7.4.1 Proficiency Testing Scheme (PTS)

Because the participants do not perform replicates, it is impossible to estimate the respective MU for each result provided in a PTS. However, at the end of the test, the organizer can obtain an estimate of the value assigned to the test material, along with its standard uncertainty. For this purpose, chapter 7 of ISO 13528 defines an additive measurement model.

Unfortunately, there are also five ways to calculate the standard uncertainty of each model component. The drafters have therefore maintained a rather loose consensus, which illustrates the wide variety of areas in which proficiency testing is organized. These different methods of calculating the MU are specific to the activity of the PTS organizer. It may also be that some portions of the material used for the test are still available, or the organizer may suggest that samples be taken. It can then be used as an IRM for validation studies or the setting up control charts. In Section 8.3.2, the procedure for checking test material homogeneity, which described in Annex B of ISO 13528, is also used to assess the sampling uncertainty.

7.4.2 Interlaboratory Studies

When laboratories are required to make replicates while participating in an interlaboratory analysis, it seems logical to be able to calculate each result provided an estimate of its MU. ISO 21748:2017 describes a solution for this situation. This standard has already been mentioned in Section 7.2 to propose a generic measurement model because applicable to any analytical method.

Measurement model

$$Z = X + \delta + L + \sum_{p=1}^P c_p G_p + E \quad (7.1)$$

The definition of the elements of this model is slightly modified in the case of an interlaboratory analysis.

Z	Output quantity.
X	Average value assigned to the test material analyzed.
δ	Intrinsic bias to the measurement method.
L	Random variable for the laboratory factor effects and no longer the series effects.
G_p	Deviations from the nominal value of the assigned value X .
c_p	Sensitivity coefficient.
E	Residual random error under repeatability conditions.

The law of propagation of uncertainty explained in Section 6.7 applies to this additive model and, at the cost of some simplifications, the standard uncertainty of the measurand is equal:

Standard variance of Z

$$u^2(Z) \approx s_L^2 + s_r^2 + u^2(\delta) \quad (7.5)$$

Variance of the bias

$$u^2(\delta) = s_{\bar{Z}}^2 \quad (7.8)$$

Standard uncertainty of

$$\begin{aligned} Z \ u(Z) &\approx \sqrt{s_L^2 + s_r^2 + u^2(\delta)} \\ u(Z) &\approx \sqrt{s_R^2 + u^2(\delta)} \end{aligned} \quad (7.6)$$

An interesting proposal of the ISO 21748 standard is to introduce a new parameter called “intra-laboratory reproducibility.” This ambiguous name is not only an oxymoron, but the idea behind it is interesting. It calculates an estimator of individual laboratory intermediate precision derived from the interlaboratory study data. The intra-laboratory reproducibility is denoted $s_{R_i}^2$, where i is the laboratory number and $s_{r_i}^2$ the repeatability variance of the laboratory. Then it gives:

Laboratory repeatability variance

$$s_{r_i}^2 = \frac{\sum_j^J (Z_{ij} - \bar{Z}_i)^2}{J - 1} \quad (7.16)$$

Table 7.8 LEAD – MU of the average of each laboratory.

Labs	$\bar{Z}_i$	$s_{r_i}^2$	$s_{R_i}^2$	$u(\bar{Z}_i)$	$UR\%(\bar{Z}_i) (\%)$
Lab 01	2.03	0.001900	0.021349	0.14611	14.4
Lab 02	1.94	0.003100	0.022549	0.15016	15.5
Lab 03	2.17	0.060933	0.080382	0.28352	26.2
Lab 04	2.43	0.006033	0.025482	0.15963	13.2
Lab 05	1.92	0.000933	0.020382	0.14277	14.8
Lab 06	1.88	0.000933	0.020382	0.14277	15.2
Lab 07	2.02	0.001033	0.020482	0.14312	14.1
Lab 08	2.02	0.003433	0.022882	0.15127	15.0
Lab 09	2.09	0.003233	0.022682	0.15061	14.4
Lab 10	1.99	0.000700	0.020149	0.14195	14.3
Lab 11	2.02	0.000400	0.019849	0.14089	13.9

Laboratory average result

$$\bar{Z}_i = \frac{\sum_j^{n_i} Z_{ij}}{n_i}$$

Intra–laboratory reproducibility variance

$$s_{R_i}^2 = s_{r_i}^2 + s_L^2 \quad (7.17)$$

Standard uncertainty of $\bar{Z}_i$

$$u(\bar{Z}_i) \approx \sqrt{s_{R_i}^2} \quad (7.18)$$

The formulas are applied to the LEAD dataset and recorded in Table 7.8. To verify the calculation, it must be remembered that $s_L^2 = 0.01945$ for the balance design when no outlying laboratory is removed.

The graphic illustration of this interlaboratory MU estimate is provided in Figure 7.5a where the horizontal bars represent the laboratory mean, and the vertical error bars the coverage interval of each mean. The individual MU is somewhat important, while it was indicated in Section 3.4.2 that two laboratories were outliers. This remark underlines the leading role of the interlaboratory variance s_L^2 in the computation of the “intra-laboratory reproducibility variance” as it is occasionally 20 times higher than the individual repeatability variance.

When the outliers (data or laboratories) are removed, MU values are strongly reduced, as shown in Figure 7.5b. The scale on the y-axis is the same on both figures to show this dramatic modification. In this case, $s_L^2 = 0.0003593$.

For a laboratory participating in an interlaboratory study, this approach is interesting as it is a good improvement of its work, but the MU obtained depends strongly on the performance of the other laboratories and must be interpreted with caution. It should be remembered that MU is specifically applicable to a measurement value obtained in one laboratory applying its operating procedure, whereas in this case, the estimation can be strongly influenced by other laboratories.

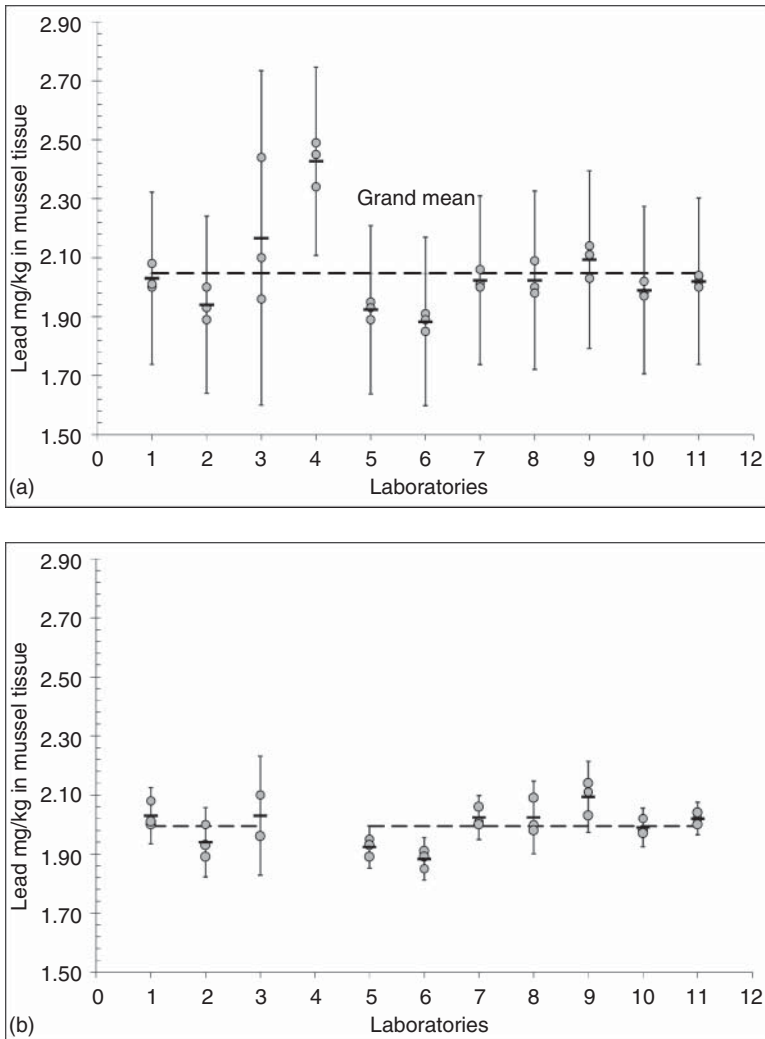


Figure 7.5 (a) LEAD – laboratory coverage intervals including outliers, (b) LEAD – individual coverage intervals after removing outliers.

7.5 Uncertainty Functions

7.5.1 Horwitz's Model

It was previously explained that MU can vary depending on the instrument or measurement method. In the case of chemical or biochemical measurements, MU also varies, sometimes very strongly, depending on the concentration level. This variation is observable, regardless of the type of measurand or concentration, estimated by inverse prediction or assigned to a validation material. In some instances, this variation can be ignored if the sample is a product of a controlled industrial process because the concentration range of the analyte may be very narrow, and the associated MU can then be taken as invariant (see Section

Usually, once an MU estimate has been obtained for a given sample concentration, it is risky to extrapolate to another sample received by the laboratory unless the concentration is close. A possible method to avoid this disadvantage is determining the dominant relationship between MU and concentration, sometimes qualified as “functional relationship.” Obviously, this requirement seems much less important for physical measurements where the same uncertainty applies to a wider range of measures.

Since the 1960s, interlaboratory studies and subsequent reproducibility calculations have been extensively used for quantitative method validation. By compiling thousands of these interlaboratory analyses, W. Horwitz demonstrated that reproducibility, i.e. precision, varies according to the concentration. The empirical model he proposed linked the relative standard deviation of reproducibility (also called the coefficient of variation of reproducibility), denoted RSD_R , to the concentration X . The basic rationale is that the RSD_R could be multiplied by 2 every time the concentration is divided by 10 [25].

Equation (7.19) presents the initial mathematical formulation of this model published by Horwitz. The result is directly given as a percentage. RSD_{TR} stands for theoretical relative standard deviation of reproducibility computed via the model while RSD_R is the observed value. The empirical model relates concentration and precision in general, whatever the method operating procedure. It is, therefore, not directly applicable to the MU of any laboratory for any method; it is rather a very general model combining all types of analytical methods.

After rearrangement, an equivalent form is given by the Eq. (7.21), which links the theoretical standard deviation of reproducibility s_{TR} to the concentration. It must be highlighted that this relationship is a power function of the same type as $f(Z) = aZ^b$. Moreover, it is a very convenient way to link these two quantities.

Horwitz model (theoretical relative standard deviation)

$$RSD_{TR} = \frac{s_{TR}}{X} \times 100 = 2^{1-0.5 \log_{10}(X)} \quad (7.19)$$

Alternative form of Horwitz model

$$RSD_{TR} = 2 \times X^{-0.155} \quad (7.20)$$

Theoretical standard deviation of reproducibility

$$s_{TR} = 0.02 \times X^{0.845} \quad (7.21)$$

Horwitz Ratio (*HorRat*)

$$HorRat = \frac{RSD_R}{RSD_{TR}} \quad (7.22)$$

A simple example illustrates the calculation of the RSD_{TR} . Consider a sample with a measured content of approximately 1 g/100 g. First, this concentration must be converted into International System units (SI), i.e. 1 g/100 g is equal to 10^{-2} kg/kg.